

Solution Requirements

Date	7 July 2026
Team ID	
Project name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	4 marks

Functional Requirements

Functional Requirements: The functional requirements define the core operations of the Smart Agricultural Production Optimization Engine (OptiCrop). The system should collect soil and weather data from users, process the information using machine learning algorithms, generate crop recommendations, suggest suitable fertilizers, and display optimization results through a user-friendly interface. These requirements describe what the system is expected to do.

Non-Functional Requirements

Non-Functional Requirements: The non-functional requirements describe the quality and performance expectations of the OptiCrop system. The application should provide fast response times, maintain reliable operation, protect user data, support future scalability, and offer an easy-to-use interface for farmers. These requirements define how well the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1.	Authentication	Secure login for users and administrators	High
2.	Authorization Levels	Admin manages data, users access prediction services	High
3.	External Interfaces	Weather API and	High

		agricultural datasets	
4.	Transaction Processing	Process user inputs and generate crop recommendations	High
5.	Reporting	Display crop prediction and fertilizer recommendation reports	Medium
6.	Business Rules	Recommend crops based on soil nutrients, pH, temperature, humidity, and rainfall	High
7.	Compliance	Protect user data and follow standard data privacy practices	Medium
8.	Other	Store prediction history for future reference	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1.	Performance & Speed Generate crop prediction quickly Response time < 2 seconds	Generate crop prediction quickly	Response time < 2 seconds
2.	Scalability Support more users and larger datasets Supports future expansion	Support more users and larger datasets	Supports future expansion
3.	Security & Data Privacy Protect user information	Protect user information	Secure login and encrypted storage

	Secure login and encrypted storage		
4.	Reliability & Availability Ensure stable system operation 99% availability	Ensure stable system operation	99% availability
5.	Usability & Accessibility Easy-to-use interface for farmers Responsive and user-friendly design	Easy-to-use interface for farmers	Responsive and user-friendly design
6.	Other Easy maintenance and future upgrades	Easy maintenance and future upgrades	Modular architecture